SSC MTS 2024 question paper with answers

18 October 2024 · Shift 2

79 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 11 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 15 questions

Q.1

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 3% per year rate of interest for 6 years.

(A) 500
(B) 480
(C) 540
(D) 580

Q.2

A sold an article to B at 40% profit and B sold it to C at 15% profit. If C paid ₹4,914 for the article, how much (in ₹, rounded off to the nearest integer) did A pay for it?

(A) 3,052
(B) 2,576
(C) 3,250
(D) 3,520

Q.4

A chef has two cans of acids and water mixture. He combines the contents of both cans in a beaker. The resulting mixture has half water and half acid. What is the quantity in which he mixed the contents of can 1 and can 2, given that can 2 has an acid-to-water ratio of 2 : 3 and can 1 has an acid-to-water ratio of 5 : 3?

(A) 3 : 7
(B) 4 : 7
(C) 4 : 5
(D) 2 : 3

Q.5

In an election, 750 votes were cast for the two contestants, Bharti and Manya. Bharti defeated Manya by 16% votes. If there were no invalid vote, then how many votes did Manya get?

(A) 430
(B) 335
(C) 120
(D) 315

Q.6

In an office, the average salary of 22 staff members was ₹81,400. If the average salary of staff and the Vice Chairman was ₹89,600, then how much is the salary of the Vice Chairman (in ₹)?

(A) 2,17,000
(B) 3,00,000
(C) 1,97,300
(D) 2,70,000

Q.7

A and B start running simultaneously, around a circular track of 1200 m, in the same direction, at a speed of 8 m/s and 12 m/s, respectively. A and B will meet at the starting point for the first time after:

(A) 300 s
(B) 150 s
(C) 100 s
(D) 80 s

Q.10

Anand wanted to make a cylindrical kaleidoscope of length 30 cm and radius 7 cm. If he is using handmade sheets to make it, then how much sheet will be required for this?

(A) 1320 cm^2
(B) 660 cm^2
(C) 1440 cm^2
(D) 1520 cm^2

Q.11

A thief is running away on a straight road on a bike moving at the speed of 28 m/s. A policeman chases him in a car moving at the speed of 30 m/s. if the instantaneous separation of the bike from the car is 400 m, how long will it take for the policeman to catch the thief?

(A) 200 seconds
(B) 100 seconds
(C) 300 seconds
(D) 150 seconds

Q.12

A shopkeeper offers buy three, get one free; what is the percentage discount being offered by the shopkeeper in this scheme?

(A) 30%
(B) 33.33%
(C) 25%
(D) 20%

Q.13

P sold an article to Q at 11% profit and Q sold it to R at 11% loss. If R paid ₹19,758 for it, then at what price (in ₹) was the article purchased by P?

(A) 22,500
(B) 19,500
(C) 20,000
(D) 19,750

Q.15

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 72 kg and 36 kg is:

(A) 52.6
(B) 50.6
(C) 51.6
(D) 49.6

Q.16

In a class, there are 30 boys and 20 girls. The average score of boys in Physics is 92, whereas the average score of girls in Physics is 83. Find the average score of the whole class in Physics.

(A) 92.6
(B) 88.4
(C) 89.3
(D) 96.2

Q.17

The simple interest on a sum of ₹4,200 at the end of 3 years is ₹756. What will be the simple interest (in ₹) on ₹5,400 at the same rate and for the same time period?

(A) 980
(B) 972
(C) 998
(D) 987

Q.18

Find the simple interest (in ₹) on ₹4000 as a sum borrowed at 4% per year rate of interest for 6 years.

(A) 920
(B) 1000
(C) 900
(D) 960

Q.20

Find the HCF of 36 and 84.

(A) 12
(B) 4
(C) 18
(D) 6

Reasoning Ability and Problem Solving 16 questions

Q.1

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some trees are blue. All blue is green. Conclusions: (I) All trees are blue. (II) Some trees are green.

(A) Only conclusion (II) follows
(B) Neither conclusion (I) not (II) follows
(C) Only conclusion (I) follows.
(D) Both conclusions (I) and (II) follows

Q.2

What will come in the place of '?' in the following equation, if '+' and '×' are interchanged and '–' and '÷' are interchanged? 41 × 1 + 390 – 10 ÷ 15 =?

(A) 55
(B) 75
(C) 65
(D) 73

Q.4

P, Q, R, S, T, U and V are sitting around a circular table facing out to the centre (but not necessarily in the same order). P is sitting second to the left of U. R is sitting third to the right of S. S is an immediate neighbour of V. V is sitting to the immediate left of P. T is not an immediate neighbour of U. How many people are sitting between P and Q when counted from the right of P?

(A) Four
(B) Three
(C) Two
(D) One

Q.5

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13, can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (4, 9, 26) (2, 8, 20)

(A) (13, 2, 30)
(B) (23, 1, 56)
(C) (5, 3, 26)
(D) (30, 5, 49)

Q.6

37 is related to 89 following a certain logic. Following the same logic, 43 is related to 101. To which of the following is 28 related, following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 78
(B) 71
(C) 55
(D) 51

Q.7

FLRX is related to KQWC in a certain way based on the English alphabetical order. In the same way, GMSY is related to LRXD. To which of the following is HNTZ related, following the same logic?

(A) MSYW
(B) MSYE
(C) MDSY
(D) MSYD

Q.8

In a certain code language, ‘GAME’ is coded as ‘8536’ and ‘MEAN’ is coded as ‘5983’. What is the code for ‘N’ in that language?

(A) 3
(B) 5
(C) 9
(D) 8

Q.9

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some bulls are rocks. Some rocks are stones. Conclusions: (I): Some bulls are stones. (II): All stones are rocks.

(A) Both conclusions (I) and (II) follow.
(B) Only conclusion (I) follows.
(C) Neither conclusion (I) nor (II) follows.
(D) Only conclusion (II) follows.

Q.10

TGKP is related to QDHM in a certain way based on the English alphabetical order. In the same way, XBWL is related to UYTI. To which of the following is VEAH related, following the same logic?

(A) SBXE
(B) SBYF
(C) RAYE
(D) RAXF

Q.11

What will come in the place of the question mark (‘?’)in the following equation, if ‘+’ and ‘×’ are interchanged and ‘÷′ and ‘–‘ are interchanged? 19 × 7 ÷ 8 × 306 – 17 = ?

(A) 44
(B) 32
(C) 41
(D) 36

Q.12

In a certain code language, ‘derive the formula’ is written as ‘le ro da’ and ‘formula to win’ is writtenas ‘ka wb ro’. How will ‘formula’ be writtenin the given language?

(A) le
(B) da
(C) ro
(D) wb

Q.13

In a certain code, ‘pen pencil eraser’ is coded as ‘dy ph mt’, ‘pencil box crayon’ is coded as ‘lp ph st’ and ‘box pen board’ is coded as ‘st bq dy’. All the codes are two-letter codes only. What is 'board' coded as?

(A) lp
(B) dy
(C) st
(D) bq

Q.15

What should come in place of the question mark (?) in the given series? 3, ?, 13, 18, 23, 28, 33

(A) 9
(B) 7
(C) 8
(D) 5

Q.16

R, S, T, D, E, F and G are sitting around a circular table facing the centre (but not necessarily in the same order). Only three people are sitting between E and T when counted from the right of E. Only two people are sitting between R and G. S is sitting to the immediate left of D. Only two people are sitting between T and R. How many people are sitting between F and S when counted from the left of F?

(A) Two
(B) Three
(C) One
(D) Four

Q.17

In a certain code language, ‘RENT’ is coded as ‘1957’ and ‘GRIM’ is coded as ‘2469’.What is the code for ‘R’ in the given code language?

(A) 9
(B) 6
(C) 1
(D) 4

Q.18

What should come in place of the question mark (?) in the given series based on the English alphabetical order? OSM, LPJ ,IMG, FJD ,?

(A) CGA
(B) BAG
(C) CAG
(D) BGA

General Awareness 25 questions

Q.1

Select the correct order of processes used for water purification in waterworks. 1. Filtration 2. Chlorination 3. Suspended impurity sedimentation 4. Solid impurity sedimentation

(A) 1-2-3-4
(B) 2-1-4-3
(C) 4-3-1-2
(D) 4-3-2-1

Q.2

Which of the following groups of ports are located on the bank of the Bay of Bengal?

(A) Paradip and New Mangalore
(B) Tuticorin and Cochin
(C) Haldia and Kandla
(D) Haldia and Paradip

Q.3

Mysore Brothers are known for playing which musical instrument?

(A) Guitar
(B) Flute
(C) Mandolin
(D) Violin

Q.4

The Betul, Hoshangabad and Chhindwara districts of Madhya Pradesh is a part of which of the following Biosphere Reserves?

(A) Agasthyamalai
(B) Pachmarhi
(C) Manas
(D) Seshachalam

Q.5

In which of the places was Arya Samaj founded?

(A) Calcutta
(B) Bombay
(C) Madras
(D) Lahore

Q.6

The Kalidas Samman award is presented by which of the following governments?

(A) Tamil Nadu
(B) Madhya Pradesh
(C) Odisha
(D) Manipur

Q.7

'Waiting for a Visa' is the autobiography of:

(A) Jawaharlal Nehru
(B) Shyama Prasad Mukherjee
(C) Vallabhbhai Patel
(D) Bhimrao Ambedkar

Q.8
Which of the following Articles of the Constitution has a provision that 'all executive actions of the Government of India shall be expressed to be taken in the name of the President'?

(A) Article 75 (1A)
(B) Article 76 (2)
(C) Article 71 (1)
(D) Article 77 (1)

Q.9
Which of the following is observed on the birthday of Dr. Sarvepalli Radhakrishnan?

(A) Students' Day
(B) Engineers' Day
(C) Teachers' Day
(D) Constitution Day

Q.10
Which financial products are commonly offered by Microfinance Institutions in India?

(A) Complex investment options
(B) Specialised services for multinational corporations
(C) Savings, credit and insurance services tailored for low-income individuals
(D) Exclusive services for high-income individuals

Q.11
What is the standard length of a cricket pitch?

(A) 20 yards
(B) 18 yards
(C) 22 yards
(D) 24 yards

Q.12
Which of the following is an example of anaerobic activity?

(A) Jumping
(B) Skipping
(C) Swimming
(D) Walking

Q.13
What is the name of the scheme launched by the Government of India in 2023 to effectively resolve pending contractual disputes?

(A) Vivad se Vishwas I
(B) Vivad se Vishwas II
(C) Vivad se Adhar II
(D) Vivad se Janadhar I

Q.14
Shuddhi movement was initiated by which of the following organisations?

(A) Paramhans Mandali
(B) Arya Samaj
(C) Satyashodak Samaj
(D) Ramakrishna Mission

Q.15
Which of the following programmes is taken up by the Government of India for improvement of female literacy rates in India?

(A) Beti Shikshan, Beti Bachao
(B) Beti Bachao, Beti Padhao
(C) Beti Jagaron, Beti Padhao
(D) Beti Sikhao, Beti Padhao

Q.16
In the year of the coronation of Akbar, who among the following was designated as the Wakil of the Mughal kingdom with the title of 'Khan-i-Khanan'?

(A) Hakim Mirza
(B) Munim Khan
(C) Bairam Khan
(D) Abdur Rahim

Q.17
The basic structure of the Indian Constitution stands on the Government of India Act, _____.

(A) 1950
(B) 1935
(C) 1942
(D) 1949

Q.18
Which of the following is the oldest dated forts in India, as first mentioned in Alexander the Great's war records, bringing it to the 4th century BCE?

(A) Kumbhalgarh Fort
(B) Kangra Fort
(C) Gwalior Fort
(D) Mehrangarh Fort

Q.19
As per the National Multidimensional Poverty Index 2021, which of the following groups of states is the poorest?

(A) Gujarat, Maharashtra, Telangana
(B) Kerala, Karnataka, Tamil Nadu
(C) Punjab, Himachal Pradesh, Uttarakhand
(D) Bihar, Uttar Pradesh, Jharkhand

Q.20
Which edition of the ODI Cricket World Cup was held in India in 2023?

(A) 14th
(B) 16th
(C) 15th
(D) 13th

Q.21
In November 2023, who was appointed as the President-elect of the Federation of Indian Chambers of Commerce and Industry (FICCI)?

(A) Anish Shah
(B) Suraj Mahindra
(C) N Chandrasekaran
(D) Adi Godrej

Q.22
Which of the following is a precious mineral?

(A) Mica
(B) Bauxite
(C) Hematite
(D) Platinum

Q.23
Which folk dance has its roots in Maharashtra?

(A) Garba
(B) Lavani
(C) Dandiya
(D) Bihu

Q.24
Foods containing fats and carbohydrates are also called _____.

(A) fermented foods
(B) body building foods
(C) roughage
(D) energy giving foods

Q.25
Ravikirti was the court poet of which ruler?

(A) Rajaraja
(B) Harshavardhana
(C) Pulakeshin II
(D) Dantidurga

English Language and Comprehension 23 questions

Q.1

Select the most appropriate form of comparative degree that can substitute the underlined segment in the given sentence. Rohit Sharma “is better than more other batsman” in the world.

(A) better than most other batsmen
(B) the best batsmen
(C) as good as the most other batsmen
(D) better batsmen

Q.2

Select the option that can be used as a one-word substitute for the given group of words. The analysis of a person's handwriting to determine his/her personality

(A) Numismatics
(B) Philology
(C) Calligraphy
(D) Graphology

Q.3

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The happy and cheerful girls / is always running towards / the ice cream cart in / front of the garden.

(A) the ice cream cart in
(B) front of the garden
(C) the happy and cheerful girls
(D) is always running towards

Q.4

Select the option that can be used as a one-word substitute for the given group of words. The branch of geography that studies mountains and their formation

(A) Mythography
(B) Orthography
(C) Orography
(D) Palaeography

Q.5

Select the INCORRECTLY spelt word.

(A) Beginer
(B) Ceremony
(C) Prescription
(D) Disappointment

Q.6

Select the most appropriate idiom to replace the underlined segment. He thinks his new plan will undoubtedly “produce the desired result” .

(A) sour grapes
(B) go pear-shaped
(C) go suck a lemon
(D) bear fruit

Q.7

Select the most appropriate option that can substitute the underlined words in the given sentence. Losing my job turned out to be “an unexpected benefit” because it pushed me to start my own business, which has been more successful than I ever imagined.

(A) a blessing in disguise
(B) a diamond in rough
(C) a golden opportunity
(D) a mixed blessing

Q.8

Select the most appropriate ANTONYM of the underlined word in the given sentence. Anupama was a very “industrious” child and submitted all her assignments on time.

(A) industrial
(B) laborious
(C) lazy
(D) intense

Q.9

Select the most appropriate synonym of the given word. Antique

(A) Modern
(B) Novel
(C) Young
(D) Ancient

Q.11

Select the most appropriate option to fill in the blanks. I bought a _____ but it is _____.

(A) cane; cain
(B) chip; cheap
(C) cain; cane
(D) cheap; chip

Q.12

Select the most appropriate form of comparative degree that can substitute the underlined segment in the given sentence. Maheshwari is three years “junior than me.”

(A) junior than any other to me
(B) junior than that of me
(C) more junior than me
(D) junior to me

Q.14

Select the grammatically correct sentence..

(A) Rukmini put a book on an table and sat down.
(B) Rukmini put the book on the table and sat down.
(C) Rukmini put the book on an table and sat down.
(D) Rukmini put an book on the table and sat down.

Q.15

Select the most appropriate ANTONYM of the underlined word in the given sentence. He was very “inquisitive” and wanted to know each and every detail.

(A) Indeterminate
(B) Uncanny
(C) Intelligent
(D) Indifferent

Q.16

Select the most appropriate synonym of the word given in brackets to fill in the blank. Many orthodox people have a ____(hostile) attitude towards the idea of women getting educated and employed.

(A) meticulous
(B) elaborate
(C) friendly
(D) belligerent

Q.17

Identify the error in the given sentence. Ajit got to work earliest than everyone else today.

(A) today
(B) everyone else
(C) earliest than
(D) Ajit got to work

Q.18

Identify the INCORRECTLY spelt word in the given sentence. Are you sure that you can’t even write a single sentnce properly?

(A) sentnce
(B) write
(C) properly
(D) single

Q.19

Select the INCORRECTLY spelt word..

(A) Music
(B) Shadow
(C) Scari
(D) Reflect

Q.20

Select the grammatically correct sentence.

(A) Rahul is very naughty as he is the most young sibling between Rama’s three brothers.
(B) Rahul is very naughty as he is the most young sibling among Rama’s three brothers.
(C) Rahul is very naughty as he is the youngest sibling among Rama’s three brothers.
(D) Rahul is very naughty as he is the youngest sibling between Rama’s three brothers.

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. The computer can become a threat to man. It can (1)_____ his survival and privacy. It also offers excellent encouragement for unethical or criminal activities. It can (2)_____ the unemployment problem as employers prefer the computer and the 'computer-controlled robots' to humans for obvious reasons. The information storage facility the computer provides has (3)_____ to collection of personal information about individuals and storage of this information in data banks by several employing agencies. If care is not exercised (4)_____ data gathering, data integrity may suffer. That is, questionable or imperfect methods may be employed; as a result, incorrect or incomplete data may be stored, and this (5)_____ become a permanent source and used for any given purpose.

Select the most appropriate option to fill in blank no. 1.

(A) endanger
(B) tranformative
(C) encourage
(D) protect

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. The computer can become a threat to man. It can (1)_____ his survival and privacy. It also offers excellent encouragement for unethical or criminal activities. It can (2)_____ the unemployment problem as employers prefer the computer and the 'computer-controlled robots' to humans for obvious reasons. The information storage facility the computer provides has (3)_____ to collection of personal information about individuals and storage of this information in data banks by several employing agencies. If care is not exercised (4)_____ data gathering, data integrity may suffer. That is, questionable or imperfect methods may be employed; as a result, incorrect or incomplete data may be stored, and this (5)_____ become a permanent source and used for any given purpose.

Select the most appropriate option to fill in blank no. 2.

(A) worsen
(B) beat
(C) control
(D) improve

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. The computer can become a threat to man. It can (1)_____ his survival and privacy. It also offers excellent encouragement for unethical or criminal activities. It can (2)_____ the unemployment problem as employers prefer the computer and the 'computer-controlled robots' to humans for obvious reasons. The information storage facility the computer provides has (3)_____ to collection of personal information about individuals and storage of this information in data banks by several employing agencies. If care is not exercised (4)_____ data gathering, data integrity may suffer. That is, questionable or imperfect methods may be employed; as a result, incorrect or incomplete data may be stored, and this (5)_____ become a permanent source and used for any given purpose.

Select the most appropriate option to fill in blank no. 3.

(A) experienced
(B) stored
(C) involved
(D) led

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. The computer can become a threat to man. It can (1)_____ his survival and privacy. It also offers excellent encouragement for unethical or criminal activities. It can (2)_____ the unemployment problem as employers prefer the computer and the 'computer-controlled robots' to humans for obvious reasons. The information storage facility the computer provides has (3)_____ to collection of personal information about individuals and storage of this information in data banks by several employing agencies. If care is not exercised (4)_____ data gathering, data integrity may suffer. That is, questionable or imperfect methods may be employed; as a result, incorrect or incomplete data may be stored, and this (5)_____ become a permanent source and used for any given purpose.

Select the most appropriate option to fill in blank no. 4.

(A) when
(B) during
(C) unless
(D) by

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. The computer can become a threat to man. It can (1)_____ his survival and privacy. It also offers excellent encouragement for unethical or criminal activities. It can (2)_____ the unemployment problem as employers prefer the computer and the 'computer-controlled robots' to humans for obvious reasons. The information storage facility the computer provides has (3)_____ to collection of personal information about individuals and storage of this information in data banks by several employing agencies. If care is not exercised (4)_____ data gathering, data integrity may suffer. That is, questionable or imperfect methods may be employed; as a result, incorrect or incomplete data may be stored, and this (5)_____ become a permanent source and used for any given purpose.

Select the most appropriate option to fill in blank no. 5.

(A) may
(B) must
(C) have to
(D) should

Answer key

Numerical and Mathematical Ability

Q.1 **(C)**	Q.2 **(A)**	Q.4 **(C)**	Q.5 **(D)**	Q.6 **(D)**	Q.7 **(A)**
Q.10 **(A)**	Q.11 **(A)**	Q.12 **(C)**	Q.13 **(C)**	Q.15 **(B)**	Q.16 **(B)**
Q.17 **(B)**	Q.18 **(D)**	Q.20 **(A)**			

Reasoning Ability and Problem Solving

Q.1 **(A)**	Q.2 **(C)**	Q.4 **(C)**	Q.5 **(A)**	Q.6 **(B)**	Q.7 **(B)**
Q.8 **(C)**	Q.9 **(C)**	Q.10 **(A)**	Q.11 **(D)**	Q.12 **(C)**	Q.13 **(D)**
Q.15 **(C)**	Q.16 **(B)**	Q.17 **(A)**	Q.18 **(A)**		

General Awareness

Q.1 **(C)**	Q.2 **(D)**	Q.3 **(D)**	Q.4 **(B)**	Q.5 **(B)**	Q.6 **(B)**
Q.7 **(D)**	Q.8 **(D)**	Q.9 **(C)**	Q.10 **(C)**	Q.11 **(C)**	Q.12 **(A)**
Q.13 **(B)**	Q.14 **(B)**	Q.15 **(B)**	Q.16 **(C)**	Q.17 **(B)**	Q.18 **(B)**
Q.19 **(D)**	Q.20 **(D)**	Q.21 **(A)**	Q.22 **(D)**	Q.23 **(B)**	Q.24 **(D)**
Q.25 **(C)**					

English Language and Comprehension

Q.1 **(A)**	Q.2 **(D)**	Q.3 **(D)**	Q.4 **(C)**	Q.5 **(A)**	Q.6 **(D)**
Q.7 **(A)**	Q.8 **(C)**	Q.9 **(D)**	Q.11 **(B)**	Q.12 **(D)**	Q.14 **(B)**
Q.15 **(D)**	Q.16 **(D)**	Q.17 **(C)**	Q.18 **(A)**	Q.19 **(C)**	Q.20 **(C)**
Q.21 **(A)**	Q.22 **(A)**	Q.23 **(D)**	Q.24 **(B)**	Q.25 **(A)**	

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (C) 540

SI = 3,000 × 3 × 6/100 = ₹540.

Q.2 · Answer (A) 3,052

A's cost × 1.4 × 1.15 = ₹4,914, so A's cost = 4,914/1.61 = ₹3,052.17, or ₹3,052.

Q.4 · Answer (C) 4 : 5

Acid is 5/8 of can 1 and 2/5 of can 2. For a mixture that is 1/2 acid, by alligation can 1 : can 2 = (1/2 – 2/5) : (5/8 – 1/2) = 1/10 : 1/8 = 4 : 5.

Q.5 · Answer (D) 315

Bharti's margin is 16% of 750 = 120 votes, so Manya got (750 – 120)/2 = 315.

Q.6 · Answer (D) 2,70,000

With the Vice Chairman the total is 23 × 89,600 = ₹20,60,800; the staff's alone is 22 × 81,400 = ₹17,90,800. The difference, ₹2,70,000, is his salary.

Q.7 · Answer (A) 300 s

A takes 1,200/8 = 150 s a lap and B 1,200/12 = 100 s. They are first at the start together after LCM(150, 100) = 300 s.

Q.10 · Answer (A) 1320 cm²

The sheet forms the curved surface: 2πrh = 2 × 22/7 × 7 × 30 = 1,320 cm².

Q.11 · Answer (A) 200 seconds

The car gains 30 – 28 = 2 m/s on the bike, so it closes the 400 m in 400/2 = 200 seconds.

Q.12 · Answer (C) 25%

The customer pays for 3 articles and gets 4, so 1 article in 4 is free: a discount of 25%.

Q.13 · Answer (C) 20,000

P's cost × 1.11 × 0.89 = ₹19,758, so P's cost = 19,758/0.9879 = ₹20,000.

Q.15 · Answer (B) 50.6

(40 + 49 + 56 + 72 + 36)/5 = 253/5 = 50.6 kg.

Q.16 · Answer (B) 88.4

(30 × 92 + 20 × 83)/50 = (2,760 + 1,660)/50 = 4,420/50 = 88.4.

Q.17 · Answer (B) 972

Rate = 756 × 100/(4,200 × 3) = 6%. On ₹5,400: 5,400 × 6 × 3/100 = ₹972.

Q.18 · Answer (D) 960

SI = 4,000 × 4 × 6/100 = ₹960.

Q.20 · Answer (A) 12

36 = 2² × 3² and 84 = 2² × 3 × 7, so the HCF is 2² × 3 = 12.

Reasoning Ability and Problem Solving

Q.1 · Answer (A) Only conclusion (II) follows

The trees that are blue are also green, so some trees are green (II). Only some trees are said to be blue, so I does not follow.

Q.2 · Answer (C) 65

After interchanging the signs: 41 + 1 × 390 ÷ 10 – 15 = 41 + 39 – 15 = 65.

Q.4 · Answer (C) Two
Going round to the left from U: R, P, V, S, T, Q (P second to U's left, V on P's left, S beside V, R third to S's right, and T away from U). Counting rightwards from P, R and U come before Q: two people.

Q.5 · Answer (A) (13, 2, 30)
(4 + 9) × 2 = 26 and (2 + 8) × 2 = 20. Likewise (13 + 2) × 2 = 30.

Q.6 · Answer (B) 71
37 × 2 + 15 = 89 and 43 × 2 + 15 = 101. Likewise 28 × 2 + 15 = 71.

Q.7 · Answer (B) MSYE
Each letter moves 5 places forward (Y wraps round to D): FLRX → KQWC and GMSY → LRXD. So HNTZ → MSYE.

Q.8 · Answer (C) 9
M, E and A are in both words, as are 3, 5 and 8 in both codes. The letter left in MEAN is N and the digit left in its code is 9, so N = 9.

Q.9 · Answer (C) Neither conclusion (I) nor (II) follows.
The rocks that are bulls need not be the rocks that are stones, so I does not follow; and 'some rocks are stones' does not make every stone a rock, so II does not follow either.

Q.10 · Answer (A) SBXE
Each letter moves 3 places back (A wraps round to X): TGKP → QDHM and XBWL → UYTI. So VEAH → SBXE.

Q.11 · Answer (D) 36
After interchanging the signs: 19 + 7 – 8 + 306 ÷ 17 = 19 + 7 – 8 + 18 = 36.

Q.12 · Answer (C) ro
'formula' is the only word in both sentences and 'ro' the only code in both, so formula = ro.

Q.13 · Answer (D) bq
'pencil' is in the first two and 'ph' in both their codes; 'pen' is in the first and third, coded 'dy'; 'box' is in the second and third, coded 'st'. That leaves 'bq' for 'board'.

Q.15 · Answer (C) 8
The terms rise by 5: 3, 8, 13, 18, 23, 28, 33.

Q.16 · Answer (B) Three
T is fourth to E's right; R, with two people between it and T, sits on E's right, and G, with two between it and R the other way round, on T's right. S and D take the seats between R and T, S on D's left: going right from E, R, S, D, T, G, F. Counting leftwards from F, G, T and D come before S: three people.

Q.17 · Answer (A) 9
R is the only letter in both words and 9 the only digit in both codes, so R = 9.

Q.18 · Answer (A) CGA
Each letter moves 3 places back: O, L, I, F, C; S, P, M, J, G; M, J, G, D, A. The next cluster is CGA.

General Awareness

Q.1 · Answer (C) 4-3-1-2
Waterworks first let the heavy solid impurities settle, then make the fine suspended ones settle (with alum, for instance), then filter the water and finally chlorinate it to kill germs.

Q.2 · Answer (D) Haldia and Paradip
Haldia (West Bengal) and Paradip (Odisha) are on the Bay of Bengal. New Mangalore, Cochin and Kandla are on the Arabian Sea, and Tuticorin on the Gulf of Mannar.

Q.3 · Answer (D) Violin
The Mysore Brothers, Mysore Nagaraj and Mysore Manjunath, are Carnatic violinists.

Q.4 · Answer (B) Pachmarhi
The Pachmarhi Biosphere Reserve, in the Satpura range, spreads over parts of the Betul, Hoshangabad and Chhindwara districts.

Q.5 · Answer (B) Bombay
Swami Dayanand Saraswati founded the Arya Samaj in Bombay in 1875.

Q.6 · Answer (B) Madhya Pradesh
The Kalidas Samman, a national award for the arts, was instituted by the Government of Madhya Pradesh in 1980.

Q.7 · Answer (D) Bhimrao Ambedkar
'Waiting for a Visa' is B. R. Ambedkar's short autobiographical account of the untouchability he faced.

Q.8 · Answer (D) Article 77 (1)
Article 77, on the conduct of business of the Government of India, says this in clause (1).

Q.9 · Answer (C) Teachers' Day
Dr Sarvepalli Radhakrishnan's birthday, 5 September, is Teachers' Day in India.

Q.10 · Answer (C) Savings, credit and insurance services tailored for low-income individuals
Microfinance institutions give small loans, savings and insurance to low-income households that ordinary banking does not reach.

Q.11 · Answer (C) 22 yards
A cricket pitch is 22 yards (20.12 m) long between the wickets.

Q.12 · Answer (A) Jumping
SSC's key gives jumping: anaerobic activity is a short, intense burst, such as a jump, in which the muscles need energy faster than oxygen can supply it. Walking and swimming at a steady pace are aerobic, and so, as a rule, is skipping, though fast skipping can be anaerobic too.

Q.13 · Answer (B) Vivad se Vishwas II
Vivad se Vishwas II (Contractual Disputes), announced in the Union Budget 2023-24, is a voluntary settlement scheme for the pending contractual disputes of the government and its undertakings.

Q.14 · Answer (B) Arya Samaj
The Arya Samaj ran the Shuddhi movement to bring back into Hinduism those who had converted to other faiths.

Q.15 · Answer (B) Beti Bachao, Beti Padhao
Beti Bachao, Beti Padhao, launched in 2015, aims to protect the girl child and see her educated.

Q.16 · Answer (C) Bairam Khan
At Akbar's accession in 1556 Bairam Khan became his regent, the vakil, with the title Khan-i-Khanan.

Q.17 · Answer (B) 1935
Much of the Constitution's structure, including the federal scheme and the office of Governor, was taken from the Government of India Act, 1935.

Q.18 · Answer (B) Kangra Fort
SSC's key gives Kangra Fort in Himachal Pradesh, which heritage and tourism accounts describe as mentioned in Alexander the Great's war records. Kumbhalgarh and Mehrangarh were built in the 15th century.

Q.19 · Answer (D) Bihar, Uttar Pradesh, Jharkhand
NITI Aayog's National Multidimensional Poverty Index 2021 found the highest shares of poor people in Bihar, Jharkhand and Uttar Pradesh.

Q.20 · Answer (D) 13th
The 2023 ODI World Cup, held in India, was the 13th edition of the tournament.

Q.21 · Answer (A) Anish Shah
Anish Shah, Group CEO of the Mahindra Group, was named FICCI's President-elect in November 2023 and took over as President in December 2023.

Q.22 · Answer (D) Platinum
Platinum is a precious metal. Mica is a non-metallic mineral, and bauxite and hematite are ores of aluminium and iron.

Q.23 · Answer (B) Lavani
Lavani is a folk dance of Maharashtra, performed to the beat of the dholki. Garba and Dandiya are from Gujarat, Bihu from Assam.

Q.24 · Answer (D) energy giving foods
Carbohydrates and fats are the body's main sources of energy, so they are called energy-giving foods; proteins are the body-building foods.

Q.25 · Answer (C) Pulakeshin II
Ravikirti, court poet of the Chalukya king Pulakeshin II, composed the Aihole inscription.

English Language and Comprehension

Q.1 · Answer (A) better than most other batsmen
Comparing one player with the rest of a group takes 'better than most other' and a plural noun. The sheet underlines 'is' as well, but no option repeats it, so the verb stays: 'Rohit Sharma is better than most other batsmen in the world.'

Q.2 · Answer (D) Graphology
Graphology is the study of handwriting as a guide to character. Calligraphy is the art of fine handwriting, numismatics the study of coins and philology the study of the history of languages.

Q.3 · Answer (D) is always running towards
The plural subject 'girls' takes a plural verb: 'are always running towards'.

Q.4 · Answer (C) Orography
Orography is the study of mountains. Orthography is spelling and palaeography the study of ancient writing.

Q.5 · Answer (A) Beginer
The correct spelling is 'beginner'.

Q.6 · Answer (D) bear fruit
To 'bear fruit' is to produce the desired result. To 'go pear-shaped' is to go wrong, and 'sour grapes' is pretending not to want what one cannot have.

Q.7 · Answer (A) a blessing in disguise
A 'blessing in disguise' is something that seems bad at first but turns out to be good: an unexpected benefit.

Q.8 · Answer (C) lazy
'Industrious' means hard-working; its opposite is 'lazy'.

Q.9 · Answer (D) Ancient
'Antique' means very old: 'ancient'.

Q.11 · Answer (B) chip; cheap
'A' needs a noun and 'it is' an adjective: 'I bought a chip but it is cheap.' 'Cheap' is not a noun, and 'cain' is not a common English word.

Q.12 · Answer (D) junior to me
'Junior' and 'senior' take 'to', not 'than': 'three years junior to me'.

Q.14 · Answer (B) Rukmini put the book on the table and sat down.
'An' goes only before a vowel sound, so 'an table' and 'an book' are wrong; only 'Rukmini put the book on the table and sat down' avoids both.

Q.15 · Answer (D) Indifferent
'Inquisitive' means eager to know; its opposite is 'indifferent', not interested.

Q.16 · Answer (D) belligerent
'Hostile' means unfriendly and aggressive: 'belligerent'.

Q.17 · Answer (C) earliest than
'Than' follows a comparative, so 'earlier than', not the superlative 'earliest'.

Q.18 · Answer (A) sentnce
The correct spelling is 'sentence'.

Q.19 · Answer (C) Scari
The correct spelling is 'scary'.

Q.20 · Answer (C) Rahul is very naughty as he is the youngest sibling among Rama's three brothers.
The superlative of 'young' is 'youngest', and a choice from three or more takes 'among', not 'between'.

Q.21 · Answer (A) endanger
A computer that is a 'threat' to man can 'endanger' his survival and privacy.

Q.22 · Answer (A) worsen
Employers preferring computers and robots to people would 'worsen' the unemployment problem.

Q.23 · Answer (D) led
'Has led to': the storage facility has resulted in personal information being collected.

Q.24 · Answer (B) during
Care must be taken 'during' data gathering, while it is going on.

Q.25 · Answer (A) may
The passage speaks of what is possible ('may suffer', 'may be stored'), so the data 'may' become a permanent source.

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